

QF620 Stochastic Modelling in Finance
Assignment 2/4
Due Date: 23-Oct-2024

1. Assume that a stock price S_t follows the Black-Scholes lognormal process:

$$dS_t = rS_t dt + \sigma S_t dW_t.$$

- (a) Use Itô's formula to derive the stochastic differential equation for the process S_t^2 .
- (b) Solve the stochastic differential equations of S_t and S_t^2 .
- (c) Use your solutions to verify the following expectations:

$$\begin{aligned}\mathbb{E}[S_t] &= S e^{rt} \\ \mathbb{E}[S_t^2] &= S^2 e^{(2r+\sigma^2)t},\end{aligned}$$

2. Digital (or binary) options are characterized by the binary state of their payoffs. A cash-or-nothing digital call option pays \$1 if $S_T > K$ on maturity, and 0 otherwise. We can write the payoff as

$$V_{\text{Cash Digital}}(T) = \mathbb{1}_{S_T > K}.$$

Using the Black-Scholes model, the stock price follows a lognormal process

$$dS_t = rS_t dt + \sigma S_t dW_t^*.$$

Derive a valuation formula for this call option:

$$V_{\text{Cash Digital}}(0) = e^{-rT} \mathbb{E}[\mathbb{1}_{S_T > K}].$$

3. An asset-or-nothing digital call option pays S_T if $S_T > K$ on maturity, and 0 otherwise. We can write the payoff as

$$V_{\text{Asset Digital}}(T) = S_T \mathbb{1}_{S_T > K}.$$

Again using the Black-Scholes model, derive a valuation formula for the asset-or-nothing digital call option:

$$V_{\text{Asset Digital}}(0) = e^{-rT} \mathbb{E}[S_T \mathbb{1}_{S_T > K}].$$

Q1

a)

Ito's formula: $df(w_t) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial w_t} dw_t + \frac{1}{2} \frac{\partial^2 f}{\partial w_t^2} d(w_t)$

$$\therefore f(S_t) = S_t^2$$

$$\therefore df(S_t) = \frac{\partial f}{\partial S_t} dS_t + \frac{1}{2} \frac{\partial^2 f}{\partial S_t^2} (dS_t)^2$$

first derivatives: $\frac{\partial f}{\partial S_t} = 2S_t$

second derivatives: $\frac{\partial^2 f}{\partial S_t^2} = 2$

$$\therefore d(S_t^2) = 2S_t dS_t + \frac{1}{2} \cdot 2 \cdot (dS_t)^2$$

$$= 2S_t dS_t + (dS_t)^2$$

$$(dS_t)^2 = (r S_t dt + \sigma S_t dW_t)^2$$

$\therefore$ The effect of the deterministic drift term is very small, so we only need to keep the stochastic diffusion term.

$$\therefore (dS_t)^2 = \sigma^2 S_t^2 (dW_t)^2$$

$$= \sigma^2 S_t^2 dt$$

$$\therefore df(S_t^2) = 2S_t (r S_t dt + \sigma S_t dW_t) + \sigma^2 S_t^2 dt$$

$$= 2r S_t^2 dt + 2\sigma S_t^2 dW_t + \sigma^2 S_t^2 dt$$

$$= S_t^2 dt (2r + \sigma^2) + 2\sigma S_t^2 dW_t$$

b)

$$dS_t = r S_t dt + \sigma S_t dW_t$$

We use $X_t = \log(S_t)$ first, and use Ito's formula to X_t .

$$dX_t = \frac{\partial f}{\partial S_t} dS_t + \frac{1}{2} \frac{\partial^2 f}{\partial S_t^2} (dS_t)^2$$

$$= \frac{1}{S_t} (r S_t dt + \sigma S_t dW_t) - \frac{1}{2} \cdot \frac{1}{S_t^2} \cdot \sigma^2 S_t^2 dt$$

$$\frac{1}{S_t} dS_t = \frac{1}{S_t} (r S_t dt + \sigma S_t dW_t)$$

$$= r dt + \sigma dW_t$$

from the previous question: $(dS_t)^2 = \sigma^2 S_t^2 dt$

$$\therefore \frac{1}{2} \left(-\frac{1}{S_t^2}\right) (dS_t)^2 = \frac{1}{2} \left(-\frac{1}{S_t^2}\right) (\sigma^2 S_t^2 dt)$$

$$= -\frac{\sigma^2}{2} dt$$

$$\therefore dX_t = r dt + \sigma dW_t - \frac{\sigma^2}{2} dt$$

$$= \left(r - \frac{\sigma^2}{2}\right) dt + \sigma dW_t$$

$$X_t = X_0 + \int_0^t \left(r - \frac{\sigma^2}{2}\right) du + \int_0^t \sigma dW_u$$

$$= X_0 + (r - \frac{\sigma^2}{2})t + \sigma W_t$$

$$\because X_0 = \log(S_0), X_T = \log(S_T)$$

$$\therefore \log(S_T) = \log(S_0) + (r - \frac{\sigma^2}{2})t + \sigma W_t$$

$$S_T = S_0 \cdot e^{(r - \frac{\sigma^2}{2})t + \sigma W_t}$$

from the previous question, we know that :

$$d(S_t^2) = S_t^2 dt (2r + \sigma^2) + 2\sigma S_t^2 dW_t$$

$$\text{let } f(x) = \log x$$

$$df(x) = \frac{\partial f}{\partial x} + \frac{1}{2} \frac{\partial^2 f}{\partial^2 x} d(x)^2$$

$$= \frac{1}{x} dx - \frac{1}{x^2} (dx)^2$$

$$\text{let } x = S_t^2$$

$$d \log S_t^2 = \frac{1}{S_t^2} dS_t^2 - \frac{1}{2S_t^4} (dS_t^2)^2$$

$$= (2r + \sigma^2) dt + 2\sigma dW_t - \frac{1}{2S_t^4} \cdot 4\sigma^2 \cdot S_t^4 dt$$

$$\int_0^T d \log S_t^2 = \int_0^T (2r - \sigma^2) dt + \int_0^T 2\sigma dW_t$$

$$\log S_T^2 - \log S_0^2 = (2r - \sigma^2)(T - 0) + 2\sigma(W_T - W_0)$$

$$\log \frac{S_T^2}{S_0^2} = (2r - \sigma^2)t + 2\sigma W_t$$

$$S_t^2 = S_0^2 \cdot e^{(2r - \sigma^2)t + 2\sigma W_t}$$

(C)

$$E(S_t) = E(S_0 \cdot e^{(r - \frac{\sigma^2}{2})t + \sigma W_t})$$

$$= S_0 \cdot e^{(r - \frac{\sigma^2}{2})t} + E(e^{\sigma W_t})$$

$$\because W_t \sim N(0, t)$$

$$= S_0 \cdot e^{(r - \frac{\sigma^2}{2})t} + e^{\frac{\sigma^2 t}{2}}$$

$$= S_0 \cdot e^{(r - \frac{\sigma^2}{2})t + \frac{\sigma^2 t}{2}}$$

$$= S_0 e^{rt - \frac{\sigma^2 t}{2} + \frac{\sigma^2 t}{2}}$$

$$= S_0 e^{rt}$$

$$E(S_t^2) = E(S_0^2 \cdot e^{(2r + \sigma^2)t + 2\sigma W_t})$$

$$= S_0^2 \cdot e^{(2r + \sigma^2)t} \cdot E(e^{2\sigma W_t})$$

$$= S_0^2 \cdot e^{(2r + \sigma^2)t} \cdot e^{2\sigma^2 t}$$

$$= S_0^2 \cdot e^{2rt + \sigma^2 t + 2\sigma^2 t}$$

$$= S_0^2 \cdot e^{2rt + 3\sigma^2 t}$$

, we don't need to add coefficient of $\sigma^2 t$.

$$= S_0^2 e^{(2r + \sigma^2)t}$$

Q2

use Black-Scholes model:

$$dS_t = rS_t dt + \sigma S_t dW_t^*$$

but we already know:

$$S_T = S_0 \cdot e^{(r - \frac{\sigma^2}{2})T + \sigma W_T^*}, W_T^* \sim N(0, T)$$

and:

$$V_{\text{cash digital}}(T) = 1_{S_T > K}, \text{ when } t = 0.$$

$$V_{\text{cash digital}}(0) = e^{-rT} \cdot E(1_{S_T > K}), \text{ so we will calculate expected value in the next step.}$$

take log of S_T :

$$\log(S_T) = \log(S_0) + (r - \frac{\sigma^2}{2})T + \sigma W_T^* > \log(K)$$

$$\sigma W_T^* > \log(K) - \log(S_0) - (r - \frac{\sigma^2}{2})T$$

$$\sigma W_T^* > \log(\frac{K}{S_0}) - (r - \frac{\sigma^2}{2})T$$

divided by σ for both side:

$$W_T^* > \frac{\log(\frac{K}{S_0}) - (r - \frac{\sigma^2}{2})T}{\sigma}$$

' $W_T^* \sim N(0, T)$, we should standardize it:

$$\frac{W_T^*}{\sqrt{T}} \sim N(0, 1)$$

$$\therefore P\left(\frac{W_T^*}{\sqrt{T}} > \frac{\log(\frac{K}{S_0}) - (r - \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}\right)$$

$$P\left(\frac{W_T^*}{\sqrt{T}} > x^*\right)$$

$$\text{let } x^* = \frac{\log(\frac{K}{S_0}) - (r - \frac{\sigma^2}{2})T}{\sigma\sqrt{T}} = - \frac{\log(\frac{S_0}{K}) + (r - \frac{\sigma^2}{2})T}{\sigma\sqrt{T}} = -d_2.$$

$$\therefore P(W_T^* > x^*) = P(Z > x^*), Z \sim N(0, 1)$$

'CDF of $Z(x)$ is $Z \sim N(0, 1)$

$$\therefore P(Z \leq x^*) = \Phi(x^*)$$

$$P(Z > x^*) = 1 - \Phi(x^*)$$

$$\begin{aligned} \therefore P(S_T > K) &= 1 - \Phi(x^*) \\ &= 1 - \Phi(-d_2) \\ &= \Phi(d_2) \end{aligned}$$

$$V_C(0) = e^{-rT} E[1_{S_T > K}]$$

$$= e^{-rT} \cdot P(S_T > K)$$

$$= e^{-rT} \cdot \Phi(d_2)$$

Q3

In Black-Scholes model, $S_T = S_0 e^{(r - \frac{1}{2}\sigma^2)T + \sigma W_T}$

$$W_T \sim N(0, T)$$

Conditional probability formula: $E(X|A) = \frac{E(X1_A)}{P(A)}$,and from Q2, we have $P(S_T > K)$

$$\therefore E(S_T | S_T > K) = E(S_T 1_{S_T > K}) \cdot P(S_T > K)$$

$$\begin{aligned} S_T > K \\ S_0 \cdot e^{(r - \frac{\sigma^2}{2})T + \sigma \sqrt{T}x} > K \\ \therefore x > \frac{\log(\frac{K}{S_0}) - (r - \frac{\sigma^2}{2})T}{\sigma\sqrt{T}} = x^* \end{aligned}$$

$$\therefore E(S_T | S_T > K)$$

$$\begin{aligned} &= \int_{x^*}^{\infty} S_T f(x) dx \\ &= \int_{x^*}^{\infty} S_0 \cdot e^{(r - \frac{\sigma^2}{2})T + \sigma\sqrt{T}x} \cdot \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}} dx \\ &= \frac{S_0 \cdot e^{(r - \frac{\sigma^2}{2})T}}{\sqrt{2\pi}} \int_{x^*}^{\infty} e^{\sigma\sqrt{T}x - \frac{x^2}{2}} dx \\ &= \frac{S_0 \cdot e^{(r - \frac{\sigma^2}{2})T}}{\sqrt{2\pi}} \int_{x^*}^{\infty} e^{\frac{2\sigma\sqrt{T}x - x^2}{2}} dx \\ &= \frac{S_0 \cdot e^{(r - \frac{\sigma^2}{2})T}}{\sqrt{2\pi}} \int_{x^*}^{\infty} e^{\frac{2\sigma\sqrt{T}x - x^2 - \sigma^2 T + \sigma^2 T}{2}} dx \\ &= \frac{S_0 \cdot e^{(r - \frac{\sigma^2}{2})T}}{\sqrt{2\pi}} \int_{x^*}^{\infty} e^{-\frac{(x - \sigma\sqrt{T})^2}{2}} \cdot e^{\frac{\sigma^2 T}{2}} dx \\ &= \frac{S_0 e^{rT}}{\sqrt{2\pi}} \int_{x^*}^{\infty} e^{-\frac{(x - \sigma\sqrt{T})^2}{2}} dx \end{aligned}$$

$$\begin{aligned} \text{let } y = x - \sigma\sqrt{T} & \quad \left| \quad \begin{aligned} \text{if } x = x^*, y = x^* - \sigma\sqrt{T} \\ x \rightarrow \infty, y \rightarrow \infty \end{aligned} \right. \\ dy = dx \end{aligned}$$

$$\begin{aligned} &= \frac{S_0 e^{rT}}{\sqrt{2\pi}} \int_{x^* - \sigma\sqrt{T}}^{\infty} e^{-\frac{y^2}{2}} dy \\ &= S_0 e^{rT} (\Phi(\infty) - \Phi(x^* - \sigma\sqrt{T})) \\ &= S_0 e^{rT} \Phi\left(\frac{\log \frac{S_0}{K} + (r + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}\right) \end{aligned}$$

$$\text{let } \frac{\log \frac{S_0}{K} + (r + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}} = d_1$$

$$= S_0 e^{rT} \Phi(d_1)$$

$$E(S_T | S_T > K) = S_0 e^{rT} \Phi(d_1) \Phi(d_2)$$

$$\therefore V_{AP}(0) = e^{rT} \cdot E(S_T | S_T > K)$$